

Drivers' Interaction Preferences in Autonomous Vehicle Multimodal Interactive Systems

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Within the scope of a broader project (which aimed to develop and implement a method for designing driver environments of autonomous vehicles), we investigated driver-vehicle interaction preferences within a multi-interactive, multi-display truck. Fifteen professional drivers explored different combinations of input (Buttons, Touch, Touchless gestures) and display (Instrument Cluster [ICL], Tablet, and Head Up Display [HUD]) and responded on the appropriateness, acceptance, and usability of the pairing. Results indicated that (1) Touchless input was not suitable for any display; (2) Buttons are best paired with the ICL and HUD; and (3) Touch was best paired with the Tablet. Future work should evaluate if the interactions with the system are easy to perform (i.e. interaction optimization); and driver understanding of system actions.

Once imagined only as science fiction, self-driving vehicles are quickly becoming reality (Levinson et al., 2011; Markoff, 2013). The Methods for Designing Autonomous Systems (MODAS; Krupenia et al., 2014) project aimed to develop and implement a method for designing the driver environment (or cabin) of highly-autonomous heavy commercial vehicles ("trucks"). The goal of the current study was to investigate methods by which future 'drivers' could interact with their self-driving truck.

In earlier work within MODAS, a series of non-interactive HUD prototypes had been created. These displays were a set of non-interactive visual displays superimposed on images of representative driving situations. In this article, we describe an investigation into how drivers would like to interact with information presented on different visual displays. This was accomplished by identifying drivers' input control preferences for interacting with the visual displays using several different input modalities. In what follows, we first present a brief background of the work conducted to construct the non-interactive HUD information display followed by the interaction study and some conclusions.

BACKGROUND

There were three primary goals to the MODAS project. First, to create a structured method for designing future (non-existent) systems that incorporate a range of advanced autonomous systems. Second, to implement this method to create a prototype cabin for a highly autonomous truck. Third, to assess this system in terms of safety, efficiency, and performance (Krupenia, et al. 2014). The method used within MODAS is based on the Goals, (Mental) Models, Observability, Control (GMOC) model (Sandbald, Andersson, Kauppi, & Wikström, 2012). The application of the GMOC model within the MODAS project is explained in detail elsewhere (Jansson, Stensson, Bodin, Axelsson, & Tschirner, 2014), however a brief summary is provided here.

To develop the non-interactive HUDs, a Work Domain Analysis (WDA) was conducted to identify the goals and constraints of the future system. Next, an observational study was performed to identify situations in which the technology would be used. This data was used as part of a Control Task

Analysis (ConTA) that examined the affordances of the available technology in the previously identified situations (Bodin, 2013). Additionally, GMOC-inspired interviews were conducted which identified the drivers' mental models. Finally, the mental models were further examined via interviews and observations (Krupenia et al., 2013).

On the basis of the information gathered, three HUD concept were developed to visualize the key findings and to provide a focus point for future discussions. Two concepts were selected for further investigation; a Mental Model based display (Figure 1), and an Operational-Tactical-Strategic based display (Figure 2). These conceptual displays were adapted to six critical situations identified during the Control Task Analysis (Bodin, 2013).

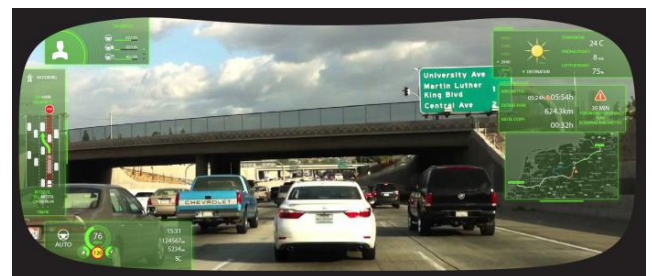


Figure 1. Mental Model Based Display for Highways.



Figure 2. Operational-Tactical-Strategic Based Display for Highway.

The design perspective taken with each of the two displays are fundamentally different. The Mental Model based display used HUD widgets to present pertinent groups of information intended to support the construction of correct

Mental Models. The Mental models represented by the widgets included: (1) immediate surroundings; (2) the truck; (3) landscape/environment; (4) general traffic; (5) cargo; (6) convoy (when relevant); and finally, because we envisaged a driver-vehicle team, (7) the truck's model of the driver.

The Operational-Tactical-Strategic based display presented information to support the supervision and control of Operational, Tactical, and Strategic tasks (Michon, 1985; Lee & Strayer, 2004). In brief, Operational tasks are behaviors occurring between 0.5 and 5 seconds (e.g. pressing/depressing the accelerator and brakes, and using controls; Schaap, van der Horst, & van Arem, 2013). Tactical tasks are behaviors occurring between 5 and 60 seconds (e.g. road following, maintaining steady speeds, and turning; Schaap et al., 2013; Michon, 1979). Finally, strategic tasks occur over minutes to days (e.g. trip goals, route, evaluating cost and benefits, planning; Michon, 1979). Using these prototypes, we could work on optimizing the design to best support the driver.

The two display concepts were adapted to six different situations and assessed as part of the MODAS 1st User Testing. The primary goals of the 1st User Testing (from the visual display perspective) was to examine the value of information content, with only a minor consideration for observability and layout of the displays. To meet the MODAS project goals (specifically, developing a prototype driver environment), further work was needed and three key activities initiated. First, to better define the information content of the displays, a Strategy Analysis was conducted that extended the work from the WDA and ConTA. Second, to create a better, more holistic (multi-modal, multi-display) environment, we examined the driver preferences for distributing the HUD information onto three possible visual displays; the Instrument Cluster, a mobile tablet device, and the HUD. Third, we investigated driver preferences for interacting with this information using three existing and/or emerging interaction technologies; steering wheel buttons, the tablet, and Touchless gestures (the focus of the current study).

At the conclusion of the 1st User Testing, there had been no investigation into how drivers would interact with the new displays. The goal of the current study was to identify the driver's Input Control preferences for interacting with information presented on the three visual displays. Three interaction technologies (Buttons, Tablet, Touchless Gestures) were paired with three display technologies (ICL, Tablet, HUD) with each input-display pairing assessed for their acceptance, usability, and 'appropriateness'.

METHOD

Participants

Data were collected from 15 professional truck drivers (2 female, 13 male) ranging in age from 23 to 59 years ($M = 41$) and with 2 to 43 years driving experience ($M = 17.2$). Testing was conducted in the Scania Truck Simulator at the Scania Technical Center (Södertälje, Sweden) and lasted about one hour each. Participants were recruited from the Scania Transport Laboratory and Scania's internal fleet of test drivers.

Inputs

The following is a description of each input modality used and their associated interaction pattern (Figure 3).

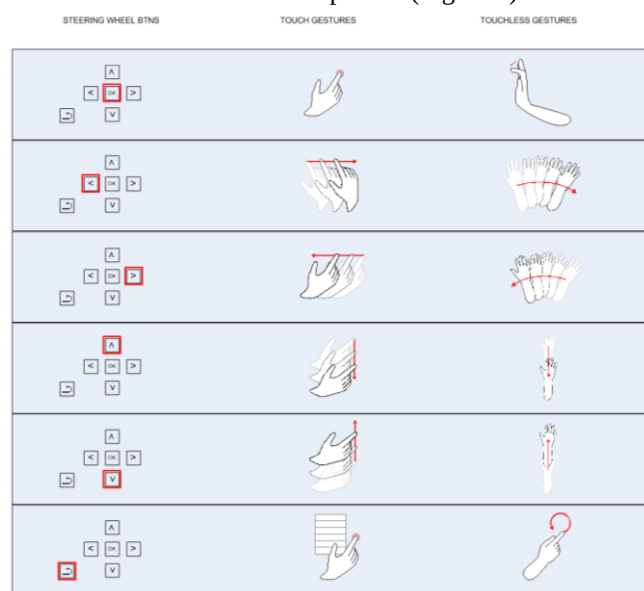


Figure 3 . Inputs and Commands for different Inputs.

The Buttons used were Racon 12 SMT/THT short travel keyswitches embedded in the right side of the steering wheel and were used for directional controls, option selection, and navigation within a hierarchical menu. Participant pressed OK to confirm selections, the left arrow to move to the left, right arrow to move to the right, up arrow to move up, down arrow to move down, and back button to move back. The Touch input was a touch screen ASUS Transformer Book 10.1" tablet which allowed the participant to interact with displays by touching areas on the tablet. The driver tapped their finger once to confirm their selection, swiped their finger right to move to the left, swiped left to move to the right, swiped down to move up, swiped up to move down, and tapped their finger away from a menu to move back. The Touchless gesture input device was a Leap Motion Dev Board v.06.5 (Figure 4)—a sensor that supports hand and finger motions as input, analogous to a mouse, but requires no contact or touching. Drivers moved their hand forward in a pressing motion to confirm a selection, swiped their hand left to move right, swiped right to move left, swiped up to move down, swiped down to move up, and drew a counterclockwise circle with their pointer finger to move back (Figure 3).



Figure 4. Touchless input, Leap Motion Dev Board v.06.5 ©.

Displays

The vehicle Instrument Cluster (ICL) is a visual display located directly in front of the driver that presents vehicle information such as fuel consumption, brake pressure, engine temperature, etc. The ICL is designed to give drivers a quick overview of all relevant truck information. The Tablet was an ASUS Transformer Book 10.1" which displays at 1280x800 in high-definition. A representative a HUD was created by projecting information from a Sony VPL-SX125 Digital Projector onto a screen directly in front of the participant—where the windshield would be from the perspective of the driver sitting in the drivers' seat.

Questionnaires

Three questionnaires were used: The Input-Output (IO) questionnaire, Acceptance Scale of Advanced Transport Telematics (AATT; van der Laan, Heino, & de Waard, 1997), and the System Usability Scale (SUS; Brooke, 1996). The IO Questionnaire measured the appropriateness of each input modality-visual display pairing by asking "Overall, how would you rate appropriateness of pairing the [Input Technology] and the [Visual Display]?" The question was answered using a five point Likert scale with higher scores indicating higher levels of appropriateness. An open ended question was used to better understand why this rating was chosen. The AATT (van der Laan, et al., 1997) assess user acceptance and is composed of two subscales, satisfaction and usefulness. The questionnaire consisted of nine items rated on a five point Likert scale. The SUS (Brooke, 1996) is an industry standard for assessing perceived system usability. The scale consists of ten items rated on a five point Likert scale, with higher scores indicating higher perceived usability.

Design

In this within-subject study, participants used every input modality (Buttons, Touch, Touchless)-visual display (ICL, Tablet, HUD) pairing while sitting in the truck simulator. Because displays and content were taken from current Scania truck systems, all participants were very familiar with the appearance and logical structure. Conditions were counterbalanced by visual display but not input control.

Procedure

Upon arrival to the simulator, participants were briefed with experiment and safety information. All participants provided informed consent prior to data collection. We also explained that we were solely interested in their expert opinion on the appropriateness of each pairing of input and display—not the specific actions (i.e. which button to press, or what Touchless gesture to make) required. During each trial, participants were asked to freely explore the system using one input modality (Buttons, Touch, and Touchless) and one output modality (ICL, Tablet, HUD). Participants were asked to think-out-loud to allow for the collection of qualitative data while interacting with the system. After each trial participants

completed the IO Questionnaire, SUS, and the AATT. After all conditions were completed, participants were asked if they had any addition comments.

RESULTS

The scores for each measure were analyzed with four separate 3 (Input) $\times$ 3 (Display) repeated measures ANOVA (IO Questionnaire, AATT Satisfaction, AATT Usefulness, and SUS). For factors that violated the assumption of sphericity the Greenhouse-Geisser correction is reported. The scores obtained (including standard error) for the Input-Output Questionnaire is presented in Table 1 and Figure 5.

Table 1. Mean scores (SEM) for Input-Output Questionnaire.

	Buttons	Touch	Touchless
ICL	4.7 $\pm$ 0.1	3.7 $\pm$ 0.4	2.5 $\pm$ 0.3
Tablet	3.3 $\pm$ 0.4	4.2 $\pm$ 0.3	2.0 $\pm$ 0.2
HUD	4.4 $\pm$ 0.2	3.7 $\pm$ 0.3	2.7 $\pm$ 0.3

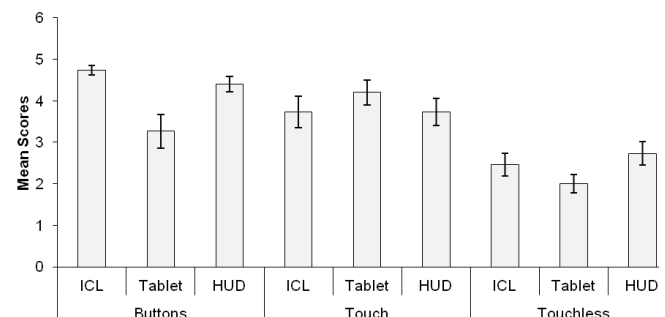


Figure 5. Input-Output Questionnaire. (Error bars = SEM)

The analysis revealed a significant main effect of Display on IO Questionnaire scores, $F(2, 28) = 5.656, p < .05$, and a significant main effect of Input on IO Questionnaire scores, $F(2, 28) = 32.718, p < 0.001$. There was a significant interaction between Input and Display on IO Questionnaire scores, $F(2, 28) = 4.218, p < .005$. Follow up comparisons of Input indicated that IO Questionnaire scores for Buttons were significantly higher than Touchless ($p < .001$) and Touch was significantly higher than Touchless ($p < .001$). Follow up comparisons of Display indicated that scores for Tablet were significantly higher than HUD ($p < .05$) and scores for ICL approach significance over the Tablet ($p = .085$). No other significant effects were found among IO Questionnaire scores.

The open comments section of the questionnaire provided an extra dimension to the data. Most participants liked the ICL-Touch pairing saying, "I am used to the system, feels natural", it was "Easy to use and does not take too much attention from the traffic", and "It is easy to handle and it goes faster. It is mobile which is an advantage.", but that it was "Hard to concentrate when touch the pad and looking at the cluster." Some participants seemed hopeful of the ICL-Touchless

pairing, but for the most part their shoulders became tired and felt uncomfortable—with some drivers asking for an armrest.

Pairing the Tablet with Buttons and Touch had a positive response. One driver mentioned it was “Good to have the hands on the steering wheel when changing menu.” The response to Tablet and Touch was that it felt natural to use and it felt “good to have everything at the same place (no buttons).” Though, participants were often frustrated with the unresponsiveness of the Tablet and Touchless pairing saying it was “Good, if it's working and you're used to it.” Many participants mentioned they liked that they could position the Tablet anywhere they liked.

The HUD elicited positive responses on its own with drivers saying they “I can keep track of the traffic at the same time as I watch the screen.” “It was easier to understand in a bigger format, good overview and you can easily see the highlighted category.” “Much easier! Its clearer where in the menu system you are and you can keep your eyes on the road instead of on the ICL.” The drivers mentioned that the HUD and Touch pairing “felt a little bit unnecessary”. The drivers also mentioned that with HUD and Touchless “there was no risk of bumping into a switch or lever” and this was the “only time it felt useful”.

Scores obtained for the AATT are presented in Figure 6.

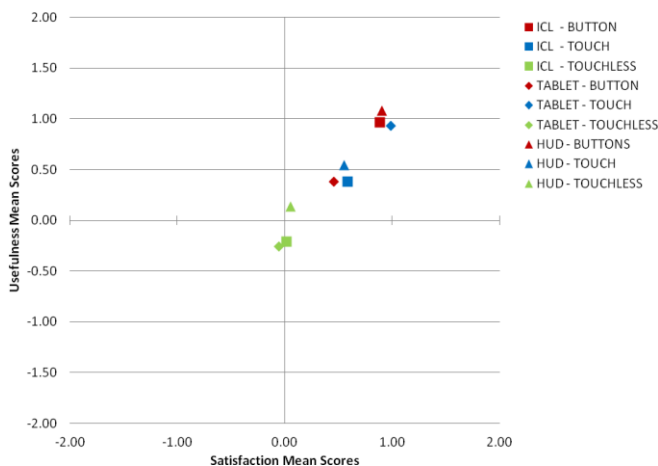


Figure 6. AATT Results.

There was an approaching significant main effect of Input on AATT Usefulness scores, $F(2, 28) = 3.57$, $p = .058$ and a significant main effect of Display, $F(2, 28) = 3.57$, $p < .001$. There was a significant interaction between Input and Display on AATT Usefulness scores, $F(2, 28) = 5.50$, $p < .01$. Follow up comparisons on Input indicated that AATT Usefulness scores for Buttons were significantly higher than Touchless ($p = .001$) and Touch was significantly higher than Touchless ($p < .01$). Follow up comparisons of Display indicated that AATT Usefulness scores for ICL scores were significantly higher than HUD ($p < .05$) and Tablet scores approached significance over HUD ($p = .052$). No other significant effects were found among AATT Usefulness scores.

There was a significant main effect of Input on AATT Satisfaction scores, $F(2, 28) = 9.349$, $p < .01$. There was no main effect of Display on AATT Satisfaction, $F(2, 28) = 0.1$,

$p = 0.894$ (and thus no follow up comparisons conducted). There was a significant interaction between Input and Display on AATT Satisfaction, $F(2, 28) = 3.104$, $p < .05$. Follow up comparisons of Display indicated that AATT Satisfaction for ICL were significantly higher than HUD ($p < .05$) and Tablet scores were significantly higher than HUD scores ($p < .01$). No other effects were found for AATT Satisfaction.

The scores obtained (including standard error) for the SUS is presented in Table 2 and Figure 7.

Table 2. SUS Mean Scores and Standard Error

	Buttons	Touch	Touchless
ICL	84.0 ± 3.3	69.3 ± 4.9	50.67 ± 5.0
Tablet	73.8 ± 3.5	82.7 ± 10.6	44.0 ± 3.8
HUD	81.67 ± 3.6	71.83 ± 14.6	48.67 ± 5.2

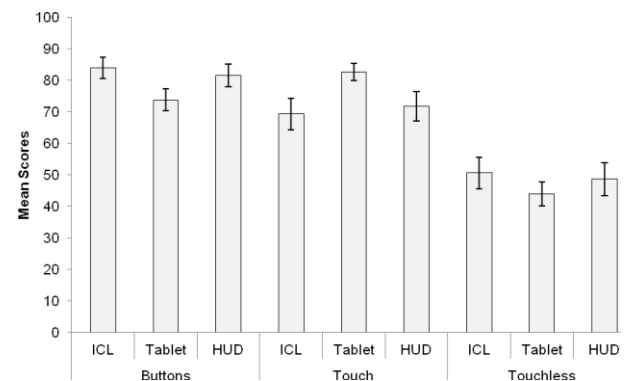


Figure 7. SUS Results. (Error bars = SEM)

There was a significant main effect of Input on SUS scores $F(2, 28) = 34.162$, $p < .01$. There was no main effect of Display on SUS scores $F(2, 28) = .087$, $p = 0.913$ (and thus no follow up comparisons were conducted). There was a significant interaction between Input and Display on SUS scores $F(2, 28) = 4.652$, $p < .01$. Follow up comparisons for Input indicated that SUS scores for Buttons were significantly higher than Touchless ($p < .01$) and Touch was significantly higher than Touchless ($p < .01$).

DISCUSSION

Comparing the results of the AATT from the Displays, both ICL and Tablet were more useful and satisfying than the HUD. This may be due to driver familiarity with the ICL and Tablet displays, but not with the HUD. When comparing the AATT Input scores, both Buttons and Touch have greater usability than Touchless. The overall Touchless results are indicative of a relatively new technology, thus there are many things yet to learn about how to design interactions with this device especially within a truck cab environment.

The initial reaction of the drivers to Touchless was that they had never seen or used anything like it. They were uncertain about how the input device worked and how to interact with the device. Coaching was necessary to instruct

the drivers on how to use the Touchless input device in order to explore the system. The Touchless gestures and interactions that we chose to control the system were not fully optimized before the study. This led to many discoveries using a trial-and-error method to get the Touchless input to work as intended while we compared our predictions with the reality of professional drivers interacting with the Touchless input device. Additionally, using the Touchless input was often frustrating because some gestures were not reliably registered by the system and the system provided no system feedback as to its current behavior. These reasons may have led to the lower scores observed across all measures involving Touchless.

The results suggested that Buttons are most useful, satisfactory, and appropriately paired with the ICL and the HUD. This may be because all of the drivers had prior experience with similar input systems while driving. The most useful and appropriate pairing for Touch is with the Tablet which is not surprising given that Touch is the native input on the tablet. Most of the drivers reported that they had prior experience or owned devices that utilized Touch input. We found that interactions with the ICL and HUD should be made with the Buttons whereas interactions with the Tablet should be made with Touch.

Despite the shortcomings of the Touchless input modality, we found that it is most appropriately paired with the HUD, but is less useful (AATT), less usable (SUS), and less appropriate (IO) compared to pairing the Buttons with the HUD. Touchless could be used with the HUD, but it should be used in combination with Buttons to give the driver an alternate input method when interacting with the HUD. Further investigation into Touchless gesture remains viable, but the current results show that the Touchless input device needs further insight, discovery, and refinement to make this a viable input device for the truck cab of the future.

In summary, we were able to identify the driver's Input Control preferences for interacting with visual information on different Visual Displays. These were used to optimize the interactions with the visual displays in the MODAS project (Krupenia et al., 2014) and potentially for the truck cab of the future.

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